

NMAM INSTITUTE OF TECHNOLOGY, NITTE
 (An Autonomous Institution affiliated to VTU, Belagavi)
Fifth Semester B.E. (E&E) (Credit System) Degree Examinations
 January 2023

20EEE262 – ELECTRIC VEHICLE BATTERY CHARGING METHODS AND TOPOLOGIES

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.*

Unit – I		Marks	BT*	CO*	PO*
1.	a) What are the major components of an electric vehicle? Also, explain the electronic drive system.	10	L*1	1	1
	b) What is meant by battery formation? Explain the effects of current density on the battery formation.	10	L2	2	1
2.	a) Explain the working principle of a fuel cell.	10	L2	1	2
	b) Write a note on the failure modes of a VRLA battery.	10	L3	2	1,2
3.	a) What are the factors effecting battery efficiency? Explain.	10	L1	1	1
	b) Write a note on battery capacity tests.	10	L2	2	1,2
Unit – II					
4.	a) Explain the multistep algorithm for charging a series string of VRLA batteries.	10	L2	3	1,2
	b) Explain the fast charging strategies of a traction battery.	10	L2	4	1
5.	a) Discuss the various charging methods of NiMH batteries.	10	L2	3	1
	b) What are the steps in calibrating the electric vehicle speedometer?	10	L1	4	1,2
6.	a) Explain the temperature based charging termination methods.	10	L2	3	1
	b) Explain the discharge characteristics of Li-ion battery.	10	L2	4	1
Unit – III					
7.	a) Discuss the model of a battery performance management system.	10	L3	5	1
	b) Write a note on battery pack safety.	10	L2	5	2
8.	a) Explain the configuration of a typical battery performance management system.	10	L2	5	1
	b) Write a note on electric vehicle component tests.	10	L1	5	2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
